

VLSI Internship – Task 1 Submission Report

Introduction to VLSI Design Flow and Basic Digital Logic Implementation

1. Objective

The objective of this task is to understand VLSI fundamentals, digital logic concepts, logic gate implementation, and combinational circuit design.

2. What is VLSI?

VLSI (Very Large Scale Integration) is the process of integrating millions or billions of transistors onto a single semiconductor chip. VLSI technology is used in processors, memory devices, SoCs, GPUs, AI accelerators, and embedded systems.

3. Applications of VLSI

- Smartphones
- Computers and Laptops
- Automotive Electronics
- Medical Devices
- AI and Machine Learning Hardware
- IoT and Embedded Systems

4. VLSI Design Flow

1. System Specification
2. Architecture Design
3. RTL Design
4. Functional Verification
5. Synthesis
6. Physical Design
7. Timing and Power Analysis
8. Fabrication
9. Testing and Validation

5. Truth Tables

AND Gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate

A	Y
0	1
1	0

NAND Gate

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

XOR Gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

6. Half Adder

Sum = A XOR B

Carry = A AND B

Half Adder Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

7. Simulation Observations

- AND gate produces output 1 only when all inputs are 1.
- OR gate produces output 1 when at least one input is 1.
- NOT gate inverts the input signal.
- NAND and NOR gates act as universal gates.
- XOR gate produces output 1 when inputs are different.
- Half Adder successfully performs one-bit binary addition using XOR and AND gates.
- All outputs matched the expected truth tables during simulation.

8. Screenshots

SIMULATION SCREENSHOTS – LOGIC GATES

1. AND GATE

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

2. OR GATE

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

3. NOT GATE

A	Y
0	1
1	0

4. NAND GATE

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

5. NOR GATE

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

6. XOR GATE

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

7. HALF ADDER

A	B	SUM (A ⊕ B)	CARRY (A · B)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

HALF ADDER EQUATIONS

SUM = A ⊕ B

CARRY = A · B

Where,

⊕ = XOR (Exclusive OR)

· = AND

OBSERVATIONS

- AND gate gives output 1 only when both inputs are 1.
- OR gate gives output 1 when any one input is 1.
- NOT gate inverts the input.
- NAND and NOR are universal gates.

- XOR gate gives output 1 when inputs are different.
- Half adder performs 1-bit addition using XOR for Sum and AND for Carry.
- All simulated results match the theoretical truth tables.